

ME 3902 — Test 3

Comparative Evaluation of Ultrasonic, Time-of-Flight, and Infrared Distance Sensors

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1. Introduction (10 pts)

This report documents Test 3, which extends the ultrasonic-and-display system built in Test 2 into a multi-sensor distance-measurement platform. Two additional distance sensors: a VL53L0X time-of-flight (ToF) sensor and an infrared (IR) detector. Were integrated alongside the existing HC-SR04 ultrasonic sensor, with all readings shown on a 128x64 OLED display and streamed to a spreadsheet for analysis. The goal was to characterize and compare three fundamentally different distance-sensing technologies: sound, laser time-of-flight, and reflected infrared, to quantify their accuracy and limitations relative to their specifications, and to evaluate how their outputs might be combined through sensor fusion.

2. System Information (50 pts)

The system is built around an Arduino Nano (ATmega328P), an 8-bit microcontroller with 2 KB of SRAM that communicates with the host PC over USB serial. It drives one output device and three input (sensing) devices:

Arduino Nano (controller). Reads all sensors, formats output, drives the display, and streams comma-separated data to CoolTerm. Its 2 KB SRAM limit proved a real design constraint (Section 6).

HC-SR04 ultrasonic sensor (input). Emits a ~40 kHz sound pulse and measures the round-trip echo time; distance is computed as $(\text{speed of sound} \times \text{time}) / 2$. It uses two digital pins (trigger and echo) and has a usable range of roughly 2-400 cm with a wide (~15 degree) detection cone.

VL53L0X time-of-flight sensor (input). Emits an invisible 940 nm laser pulse and times the photons' round trip using a single-photon avalanche diode (SPAD) array. It communicates over I2C at address 0x29, resolves to the millimeter, and ranges from about 3 cm to 1.2 m. Its beam is far narrower than the ultrasonic cone.

Infrared detector (input). A digital obstacle module (IR LED plus photodiode and comparator) with a potentiometer-set threshold. It outputs a single digital signal (LOW = object detected) rather than a continuous distance, so it is characterized by the distance at which its output switches.

SSD1306-class OLED display (output). A 128x64 monochrome panel on the I2C bus at address 0x3C, used to show all three live readings.

Because both the OLED and the ToF sensor are I2C devices with distinct addresses (0x3C and 0x29), they share the same two-wire bus (pins A4/A5) without conflict — a key architectural feature of the design.

3. Experiment Methodology (50 pts)

Circuit. The OLED and VL53L0X share the I2C bus (SDA to A4, SCL to A5). The ultrasonic sensor uses digital pins D3 (trigger) and D4 (echo); the IR detector's output connects to D2. All modules draw 5 V and ground from the breadboard power rails, fed from the Nano's 5V and GND pins. The connection map is given in Table 1.

Module	Module pin	Arduino Nano pin
HC-SR04	VCC / GND	5V rail / GND rail
	Trig / Echo	D3 / D4
VL53L0X	VIN / GND	5V rail / GND rail

Module	Module pin	Arduino Nano pin
	SDA / SCL	A4 / A5
OLED	VCC / GND	5V rail / GND rail
	SDA / SCL	A4 / A5 (shared with ToF)
IR detector	VCC / GND / OUT	5V rail / GND rail / D2

Table 1. Wiring map for the combined system.

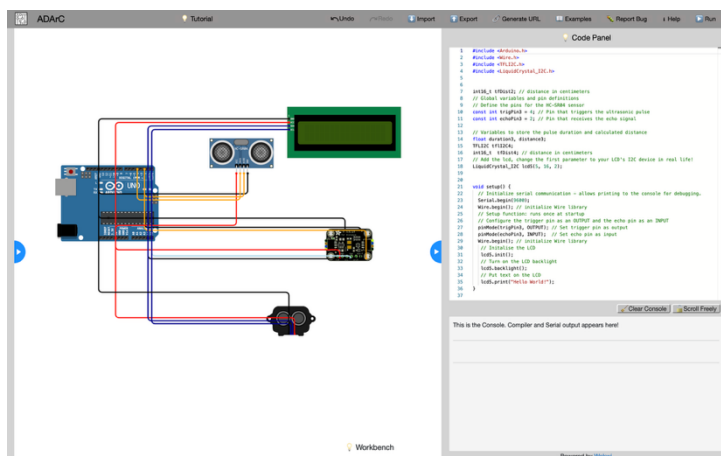
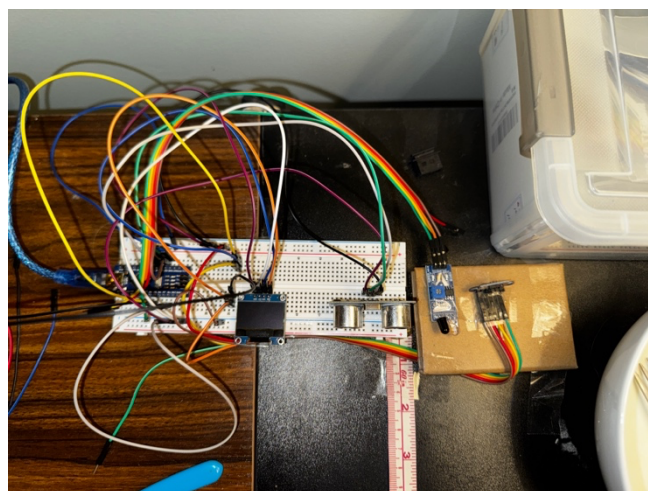


Figure 2 (Right) — Hardware photograph; components listed left to right: Arduino Nano, OLED display, Ultrasonic sensor, IR detector, Time of Flight sensor.

Figure 1 (Left) — circuit schematic (OLED display substituted for LCD1602 due to unavailability)



A single sketch reads all three sensors each loop, displays them on the OLED, and prints one CSV line per cycle (Time_ms, Ultrasonic_cm, ToF_cm, IR_detected) at 115200 baud, with a 100 ms loop delay (~10 samples/s). Out-of-range conditions are reported as -1. A notable implementation decision: the Adafruit OLED driver allocates a 1 KB frame buffer in RAM, which together with the VL53L0X library exceeded the Nano's 2 KB SRAM and caused the program to crash on start. The display was therefore driven with a buffer-less text library (U8x8), which fits within the available memory.

Each sensor was tested against known distances set with a ruler and tape measure; three readings were taken per condition and averaged to reduce noise. Tests covered a range/linearity sweep against a flat target and a material comparison at a fixed distance using an aluminum tin, clear plastic bag, glass, and clothing fabric. Because the IR detector is a threshold device, it was characterized by the trigger distance at which its output flipped for each surface, having been calibrated to trigger at 30cm on the aluminum tin.

The serial CSV stream was captured to a text file with CoolTerm, then imported into Excel (Data > From Text/CSV, comma-delimited). Out-of-range sentinel values (-1) were blanked before computing statistics so they would not bias the averages.

4. Data from Sensors (50 pts)

Measured distance was compared to true distance for the ultrasonic and ToF sensors; error is computed as (measured - actual) and percent error as (measured - actual)/actual. The IR detector is reported as its trigger distance per surface. Full per-trial data is in the accompanying Excel workbook.

Actual (cm)	US mean (cm)	US error %	ToF mean (cm)	ToF error %
3	2.3	-23.00	3.0	0.00
5	3.9	-22.00	4.7	-6.00
7.5	5.9	-21.33	7.1	-5.33

Actual (cm)	US mean (cm)	US error %	ToF mean (cm)	ToF error %
10	7.8	-22.00	9.2	-8.00
15	13.5	-10.00	14.0	-6.22
20	18.4	-8.00	18.9	-5.50
30	26.7	-11.00	30.2	0.60

Table 2. Range/linearity: measured vs. actual distance, flat target, 0 degrees, aluminum Altoids tin used as target.

Material (at 30 cm)	US mean (cm)	ToF mean (cm)	IR trigger (cm)
Altoids tin	28.3	30.2	Calibrated @ 30
White index card	28.5	31.2	29
Clear plastic bag	29.7	43.7	14
Glass cutting board	27.7	37.6	9
T-shirt	32.4	33.5	11.5

Table 3. Material comparison at a fixed 30 cm distance. (Angular-sweep data: see Excel workbook.)

5. Applications (50 pts)

These four devices (three distance sensors plus a display) can be combined into several practical systems. One worked example is given below as a model

Example — automated parking / proximity assistant. The sensors monitor the gap to an approaching object and the OLED shows the live distance, while a buffer zone triggers a warning. The ultrasonic provides robust mid-range coverage, the ToF adds precise close-range readings (down to a few centimeters), and the IR detector acts as a simple last-line presence trigger. Circuit changes would be minimal: adding a buzzer or LED on a spare digital pin. Code changes would add a threshold comparison that activates the alert when distance gets too close to an obstacle

6. Discussion and Conclusion (30 pts)

Sensor performance. The three sensors are expected to disagree in revealing ways because each is blind to something different. The ultrasonic sensor reads hard, flat surfaces well but degrades sharply on absorptive materials (foam, fabric) and at angles beyond roughly 10-15 degrees, where the echo deflects away from the receiver. The ToF sensor is highly precise on opaque surfaces but is defeated by transparent targets (glass, clear plastic), which the laser passes through, and by dark, absorptive surfaces that return too few photons; it also has a short-range floor near 3-4 cm. The IR detector is dominated by surface reflectivity, triggering much closer on dark surfaces than light ones.

This was reflected in my data, where the opaque, solid objects like the aluminum tin and paper index card measurements remained close across all the sensors. This quickly changed with translucent and transparent items like the plastic bag and cutting board wildly altering the effectiveness of the ToF and IR sensors due to their difficulty accurately measuring those materials.

Overall, the data shows the sensor with the least tendency to skew measurements due to changing conditions is the ultrasonic sensor. However, this baseline consistency is accompanied by a high error rate, particularly at distances between 5-10 cm. The ToF sensor was far more accurate when tested against the ultrasonic sensor, but it performed poorly with transparent materials, preventing the laser from accurately reflecting back to the sensor. The IR sensor worked very well for the condition it was calibrated for, reading within ~1cm of the actual measured value on flat, opaque materials. However, when conditions changed, it performed far worse due to not being adjusted for that scenario on the on-chip voltage adjuster.

Errors fall into three classes. Systematic (bias) errors include the measurement origin: the ToF measures from its lens window, so referencing the tape to the board edge introduces a constant offset. Random errors appear as sample-to-sample scatter, quantified by the standard deviation of repeated readings. Environmental errors include temperature (which changes the speed of sound and therefore the ultrasonic constant 0.0343 cm/us) and ambient infrared (which interferes with the ToF and IR sensors). Corrections follow the error's shape: a constant offset is removed by adjusting the origin or constant, while a distance-dependent error is corrected by fitting a calibration trend line to the measured-vs-actual data and using that equation in the code.

Integration challenges. Bringing the sensors together surfaced engineering issues absent from the standalone tests. Software faults split into compile/link errors (a missing object declaration; a missing loop function) and runtime faults that were really wiring problems (an I2C boot failure resolved by confirming SDA/SCL on A4/A5). The most instructive was the SRAM limit: two libraries that each worked alone overflowed memory together, forcing the buffer-less display driver — a concrete example of a resource constraint driving a design change.

Sensor fusion. Rather than trusting a single sensor, the three readings can be combined into one estimate by weighting each according to its confidence in the present conditions: discarding out-of-range values, preferring the ToF at short range where it is most precise, and down-weighting it on transparent targets where the ultrasonic is more reliable. A simple validity-weighted average, or a Kalman-style filter for moving targets, yields a single distance more robust than any one sensor alone.

7. References (5 pts)

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- [5] O. Kraus, "U8g2: Monochrome graphics and text library for embedded devices," GitHub repository. [Online].
- [6] C. Forino, "Detecting obstacle with IR Sensor and Arduino," 28 December 2018. [Online]. Available: <https://playembedded.org/blog/detecting-obstacle-with-ir-sensor-and-arduino/>. [Accessed 14 June 2026].
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- [8] SunFounder, "Lesson 21: Time of Flight Micro-LIDAR Distance Sensor (VL53L0X)," SunFounder, [Online]. Available: https://docs.sunfounder.com/projects/umsk/en/latest/02_arduino/uno_lesson21_vl53l0x.html. [Accessed 14 June 2026].

8. GitHub Code Link (20 pts)

https://github.com/CharlieJSmith57/ME-3902-E1-2026/tree/cbb0dace43cf6ec9460875f2a3eaad26b31560da/Test_3